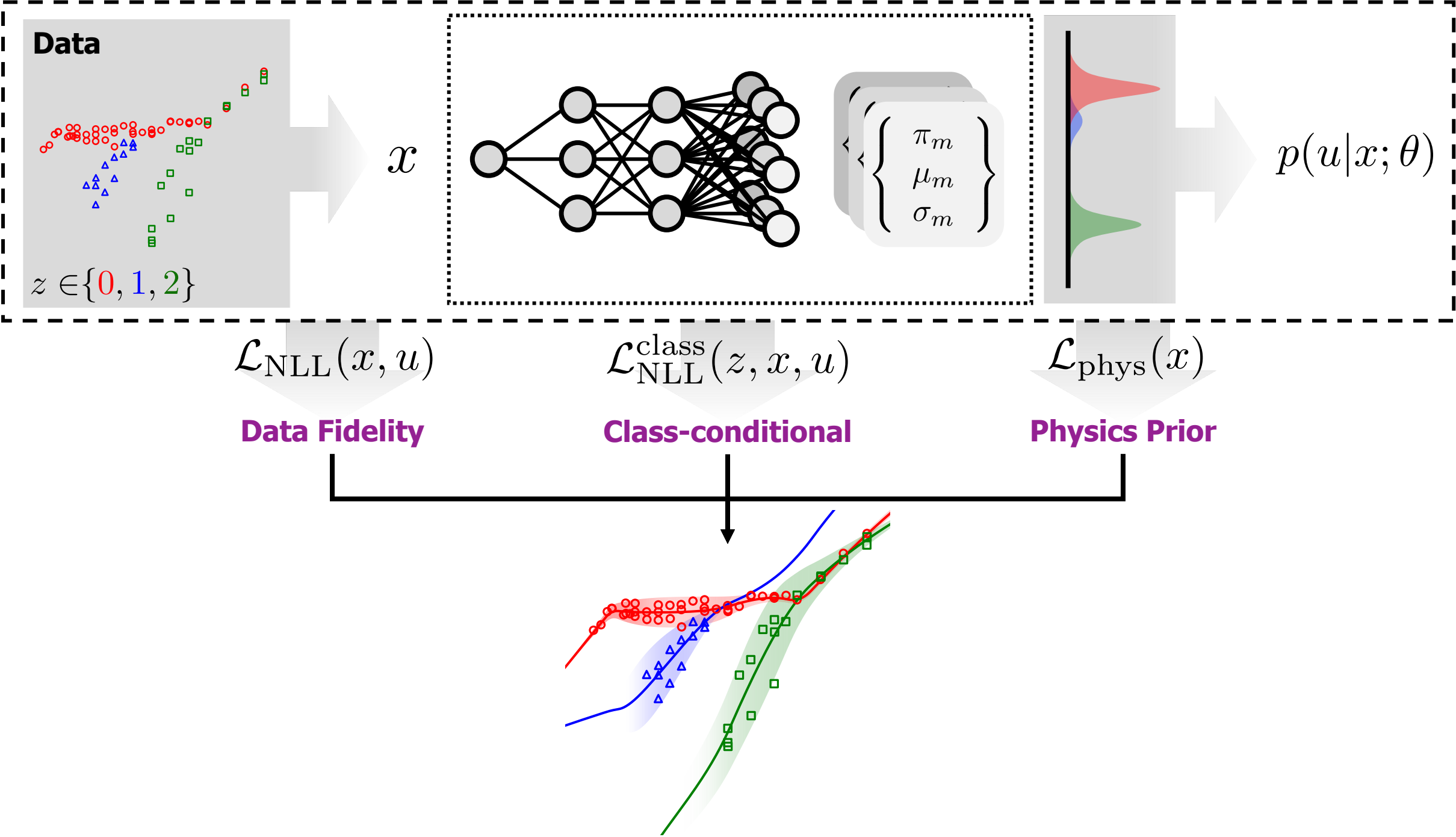

Data
$z \in \{0, 1, 2\}$
x
π_m
μ_m
σ_m
$p(u|x;\theta)$
$\mathcal{L}_{\mathrm{NLL}}(x, u)$
$\mathcal{L}^{\mathrm{class}}_{\mathrm{NLL}}(z, x, u)$
$\mathcal{L}_{\mathrm{phys}}(x)$
Data Fidelity
Class-conditional
Physics Prior